Chapter 16

Campbell Biology, AP Edition, 12e

Image of D N A double helix, which resembles a twisted ladder.

Chapter 16

The Molecular Basis of Inheritance

Figure 16.1

The elegant double-helical structure of deoxyribonucleic acid (DNA) shook the scientific world when it was proposed in April 1953 by James Watson and Francis Crick. The DNA you inherited from your parents contains all your genes—your genetic information.

How does D N A replication transmit genetic information?

Begin with an unduplicated chromosome (one D N A molecule and proteins). Take a D N A segment from a chromosome. Each gene is a unit of hereditary information consisting of a specific D N A sequence. D N A replication begins at multiple sites (origins), each forming a replication bubble with a fork at each end. D N A replication and chromosome condensation is completed. Duplicated and condensed chromosomes (two D N A molecules and proteins) form. Two D N A molecules, which are distributed to daughter cells, are formed.

Key Concepts

- **16.1DNA is the genetic material****
- **16.2Many proteins work together in DNA replication and repair****
- **16.3A chromosome consists of a DNA molecule packed together with proteins****

Advanced placement edition

The evidence that DNA and RNA carry genetic information in all life-forms **(BIG IDEA 1)** began with a series of historical experiments **(BIG IDEA 3)** that led to an understanding of how the structure and interaction of molecules **(BIG IDEA 4)** facilitate the replication and transfer of DNA and RNA across membranes **(BIG IDEA 3)**.

Study Tip

Draw DNA replication: Make up a single-stranded DNA sequence that is 40 nucleotides in length. As you go through Concepts 16.1 and 16.2, use your sequence to draw a double-stranded DNA molecule, labeling the ends. Then diagram the process of replication of your DNA molecule.

A drawing of D N A structure and replication begins with a single strand with a sequence that uses T, C, A, and G. The sequence is T C A G C G T A A G G T G A T G T A T A.

Get Ready for This Chapter

In this chapter, you will learn about the experiments that identified DNA as the genetic material, the structure of DNA and how it was discovered, how DNA is replicated, and how it is arranged in a chromosome and in the nucleus. Before beginning this chapter, you should be able to do the following:

- Describe the structure of DNA (see Concept 5.5 and Figure 5.24a).
- Describe how enzymes carry out reactions (see Concept 8.4 and Figure 8.16).
- Explain what happens to chromosomes during the cell cycle (see Concept 12.2 and Figure 12.6).
- Describe what happens to chromosomes during mitosis (see Figure 12.7).

The following questions provide a quick check of your basic knowledge in these areas.

1. The two strands of a DNA double helix are held together by _____ between pairs of nitrogenous bases.

```
ionic bonds
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That's incorrect. Try again.

Incorrect. Review the following material: DNA molecules have two polynucleotides, or “strands,” that wind around an imaginary axis, forming a double helix. The sugar-phosphate backbones are on the outside of the

helix, and the nitrogenous bases are paired in the interior of the helix. The two strands of DNA are held together by hydrogen bonds between the paired bases.

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hydrogen bonds
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That's correct.

Correct! If you had trouble with this question, review the following material: DNA molecules have two polynucleotides, or “strands,” that wind around an imaginary axis, forming a double helix. The sugar-phosphate backbones are on the outside of the helix, and the nitrogenous bases are paired in the interior of the helix. The two strands of DNA are held together by hydrogen bonds between the paired bases.

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disulfide bonds
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That's incorrect. Try again.

Incorrect. Review the following material: DNA molecules have two polynucleotides, or “strands,” that wind around an imaginary axis, forming a double helix. The sugar-phosphate backbones are on the outside of the helix, and the nitrogenous bases are paired in the interior of the helix. The two strands of DNA are held together by hydrogen bonds between the paired bases.

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covalent bonds
```

That's incorrect. Try again.

Incorrect. Review the following material: DNA molecules have two polynucleotides, or “strands,” that wind around an imaginary axis, forming a double helix. The sugar-phosphate backbones are on the outside of the helix, and the nitrogenous bases are paired in the interior of the helix. The two strands of DNA are held together by hydrogen bonds between the paired bases.

2. As a result of its involvement in a chemical reaction, an enzyme

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is unchanged by the reactants or products.
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That's correct.

Correct! If you had trouble with this question, review the following material: An enzyme is a macromolecule that acts as a catalyst, a chemical agent that speeds up a reaction without being consumed or changed by the reaction. An enzyme catalyzes a reaction by lowering the activation energy barrier, enabling

the reactant molecules to absorb enough energy to reach the transition state. Enzymes emerge from the reaction in their original form, having neither donated nor received matter or energy from the reactants.

```
        donates electrons to its reactants, turning them into
products.
```

That's incorrect. Try again.

Incorrect. Review the following material: An enzyme is a macromolecule that acts as a catalyst, a chemical agent that speeds up a reaction without being consumed or changed by the reaction. An enzyme catalyzes a reaction by lowering the activation energy barrier, enabling the reactant molecules to absorb enough energy to reach the transition state. Enzymes emerge from the reaction in their original form, having neither donated nor received matter or energy from the reactants.

```
        donates amino acid R groups to its reactants, turning
them into products.
```

That's incorrect. Try again.

Incorrect. Review the following material: An enzyme is a macromolecule that acts as a catalyst, a chemical agent that speeds up a reaction without being consumed or changed by the reaction. An enzyme catalyzes a reaction by lowering the activation energy barrier, enabling the reactant molecules to absorb enough energy to reach the transition state. Enzymes emerge from the reaction in their original form, having neither donated nor received matter or energy from the reactants.

```
        receives functional groups from reactants, allowing them
to become products.
```

That's incorrect. Try again.

Incorrect. Review the following material: An enzyme is a macromolecule that acts as a catalyst, a chemical agent that speeds up a reaction without being consumed or changed by the reaction. An enzyme catalyzes a reaction by lowering the activation energy barrier, enabling the reactant molecules to absorb enough energy to reach the transition state. Enzymes emerge from the reaction in their original form, having neither donated nor received matter or energy from the reactants.

3. During _____ of the cell cycle, the cell grows and replicates both its organelles and its chromosomes.

```
prophase
```

That's incorrect. Try again.

Incorrect. Review the following material: Interphase can be divided into three phases: G1 phase, S phase, and G2 phase. During all three phases of interphase, a cell grows by producing proteins and cytoplasmic organelles. Duplication of the chromosomes, crucial for eventual division of the cell, occurs entirely during the S phase of interphase. Cell division occurs during M phase, which can be divided into two stages: mitosis and cytokinesis.

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mitosis
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That's incorrect. Try again.

Incorrect. Review the following material: Interphase can be divided into three phases: G1 phase, S phase, and G2 phase. During all three phases of interphase, a cell grows by producing proteins and cytoplasmic organelles. Duplication of the chromosomes, crucial for eventual division of the cell, occurs entirely during the S phase of interphase. Cell division occurs during M phase, which can be divided into two stages: mitosis and cytokinesis.

```
interphase
```

That's correct.

Correct! If you had trouble with this question, review the following material: Interphase can be divided into three phases: G1 phase, S phase, and G2 phase. During all three phases of interphase, a cell grows by producing proteins and cytoplasmic organelles. Duplication of the chromosomes, crucial for eventual division of the cell, occurs entirely during the S phase of interphase. Cell division occurs during M phase, which can be divided into two stages: mitosis and cytokinesis.

```
cytokinesis
```

That's incorrect. Try again.

Incorrect. Review the following material: Interphase can be divided into three phases: G1phase, S phase, and G2 phase. During all three phases of interphase, a cell grows by producing proteins and cytoplasmic organelles. Duplication of the chromosomes, crucial for eventual division of the cell, occurs entirely during

the S phase of interphase. Cell division occurs during M phase, which can be divided into two stages: mitosis and cytokinesis.

4. During mitosis, centromeres separate and chromatids become individual chromosomes during which phase?

```
prometaphase
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That's incorrect. Try again.

Incorrect. Review the following material: During prophase and prometaphase, the duplicated chromosomes become condensed. During metaphase, the duplicated chromosomes are all aligned at the metaphase plate, a plane that is equidistant between the spindle's two poles. Anaphase is the phase where the two sister chromatids of each pair separate and each chromatid becomes an independent chromosome. This is followed by telophase, during which two daughter nuclei form in the cell, the nucleoli reappear, and the chromosomes become less condensed.

```
metaphase
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That's incorrect. Try again.

Incorrect. Review the following material: During prophase and prometaphase, the duplicated chromosomes become condensed. During metaphase, the duplicated chromosomes are all aligned at the metaphase plate, a plane that is equidistant between the spindle's two poles. Anaphase is the phase where the two sister chromatids of each pair separate and each chromatid becomes an independent chromosome. This is followed by telophase, during which two daughter nuclei form in the cell, the nucleoli reappear, and the chromosomes become less condensed.

```
anaphase
```

That's correct.

Correct! If you had trouble with this question, review the following material: During prophase and prometaphase, the duplicated chromosomes become condensed. During metaphase, the duplicated chromosomes are all aligned at the metaphase plate, a plane that is equidistant between the spindle's two poles. Anaphase is the phase where the two sister chromatids of each pair separate and each chromatid

becomes an independent chromosome. This is followed by telophase, during which two daughter nuclei form in the cell, the nucleoli reappear, and the chromosomes become less condensed.

```
telophase
```

That's incorrect. Try again.

Incorrect. Review the following material: During prophase and prometaphase, the duplicated chromosomes become condensed. During metaphase, the duplicated chromosomes are all aligned at the metaphase plate, a plane that is equidistant between the spindle's two poles. Anaphase is the phase where the two sister chromatids of each pair separate and each chromatid becomes an independent chromosome. This is followed by telophase, during which two daughter nuclei form in the cell, the nucleoli reappear, and the chromosomes become less condensed.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 16.1 Describe the structure of DNA and evidence that it is the genetic material.
- 16.2 Explain how the Meselson-Stahl experiment contributed to our understanding of DNA replication, and describe this process.
- 16.3 Distinguish among the levels of chromatin packing in a eukaryotic chromosome.

Concept 16.1: DNA is the genetic material

Today, even schoolchildren have heard of DNA, and scientists routinely manipulate DNA in the laboratory. Early in the 20th century, however, identifying the molecules of inheritance posed a major challenge to biologists.

The Search for the Genetic Material: *Scientific Inquiry*

Once T. H. Morgan's group showed that genes exist as parts of chromosomes (described in Concept 15.1), the two chemical components of chromosomes—DNA and protein—emerged as the leading candidates for the genetic material. Until the 1940s, the case for proteins seemed stronger: Biochemists had identified proteins

as a class of macromolecules with great heterogeneity and specificity of function, essential requirements for the hereditary material. Moreover, little was known about nucleic acids, whose physical and chemical properties seemed far too uniform to account for the multitude of specific inherited traits exhibited by every organism. This view gradually changed as the role of DNA in heredity was worked out in studies of bacteria and the viruses that infect them, systems far simpler than fruit flies or humans. Let's trace the search for the genetic material as a case study in scientific inquiry.

Evidence That DNA Can Transform Bacteria

In 1928, a British medical officer named Frederick Griffith was trying to develop a vaccine against pneumonia. He was studying *Streptococcus pneumoniae*, a bacterium that causes pneumonia in mammals. Griffith had two strains (varieties) of the bacterium, one pathogenic (disease-causing) and one nonpathogenic (harmless). He was surprised to find that when he killed the pathogenic bacteria with heat and then mixed the cell remains with living bacteria of the nonpathogenic strain, some of the living cells became pathogenic **(Figure 16.2)**. Furthermore, this newly acquired trait of pathogenicity was inherited by all the descendants of the transformed bacteria. Apparently, some chemical component of the dead pathogenic cells caused this heritable change, although the identity of the substance was not known. Griffith called the phenomenon ******transformation**, now defined as a change in genotype and phenotype due to the assimilation of external DNA by a cell. Later work by Oswald Avery, Maclyn McCarty, and Colin MacLeod identified the transforming substance as DNA.

Figure 16.2

Inquiry: Can a genetic trait be transferred between different bacterial strains?

Experiment

Frederick Griffith studied two strains of the bacterium *Streptococcus pneumoniae*. The S (smooth) strain can cause pneumonia in mice; it is pathogenic because the cells have an outer capsule that protects them from an animal's immune system. Cells of the R (rough) strain lack a capsule and are nonpathogenic. To test for the trait of pathogenicity, Griffith injected mice with the two strains:

Can a genetic trait be transferred between different bacterial strains?

The experiment has living S cells, living R cells, heat killed S cells, and a mixture of heat killed S cells and living R cells that are all injected into mice.

Living S cells with outer capsule (pathogenic control) injected into mouse. Result: Mouse dies.

Living R cells with no outer capsule (nonpathogenic control) injected into mouse. Result: mouse healthy.

Heat killed S cells in capsule (nonpathogenic control) injected into mouse. Result: Mouse healthy.

Mixture of heat killed S cells in capsule and living R cells without capsule injected into mouse. Result: mouse dies, and living S cells in capsules were found in mouse's blood. These cells could reproduce, producing more S cells.

Results

The result of an experiment on whether a genetic trait can be transferred between different bacterial strains.

The experiment has living S cells, living R cells, heat killed S cells, and a mixture of heat killed S cells and living R cells that are all injected into mice.

Living S cells with outer capsule (pathogenic control) injected into mouse. Result: Mouse dies.

Living R cells with no outer capsule (nonpathogenic control) injected into mouse. Result: mouse healthy.

Heat killed S cells in capsule (nonpathogenic control) injected into mouse. Result: Mouse healthy.

Mixture of heat killed S cells in capsule and living R cells without capsule injected into mouse. Result: mouse dies, and living S cells in capsules were found in mouse's blood. These cells could reproduce, producing more S cells.

Conclusion

The living R bacteria had been transformed into pathogenic S bacteria by an unknown, heritable substance from the dead S cells that enabled the R cells to make capsules.

Data from F. Griffith, The significance of pneumococcal types, *Journal of Hygiene* 27:113–159 (1928).

WHAT IF? *How did this experiment rule out the possibility that the R cells simply used the dead S cells' capsules to become pathogenic?*

*For suggested answer, see*Appendix A.**

Scientists remained skeptical, however, since many still viewed proteins as better candidates for the genetic material. Also, many biologists were not convinced that bacterial genes would be similar in composition and function to those of more complex organisms. But the major reason for the continued doubt was that so little was known about DNA.

Evidence That Viral DNA Can Program Cells

Additional evidence that DNA was the genetic material came from studies of viruses that infect bacteria **(Figure 16.3)**. These viruses are called **bacteriophages** (meaning "bacteria-eaters"), or **phages** for short. Viruses are much simpler than cells. A **virus** is little more than DNA (or sometimes RNA) enclosed by a protective coat, which is often simply protein. To produce more viruses, a virus must infect a cell and take over the cell's metabolic machinery.

Figure 16.3

A virus infecting a bacterial cell.

Phage head with D N A inside, above tail sheath and tail fibers, injecting genetic material into a bacterial cell. Scale: 100 nanometers.

A phage called T2 attaches to a host cell and injects its genetic material through the plasma membrane, while the head and tail parts remain on the outer bacterial surface (colorized TEM).

Phages have been widely used as tools by researchers in molecular genetics. In 1952, Alfred Hershey and Martha Chase performed experiments showing that DNA is the genetic material of a phage known as T2. This is one of many phages that infect *Escherichia coli* (*E. coli*), a bacterium that normally lives in the intestines of mammals and is a model organism for molecular biologists. At that time, biologists already knew that T2, like many other phages, was composed almost entirely of DNA and protein. They also knew that the T2 phage could quickly turn an *E. coli* cell into a T2-producing factory that released many copies of new phages when the cell ruptured. Somehow, T2 could reprogram its host cell to produce viruses. But which viral component—protein or DNA—was responsible?

Hershey and Chase answered this question by devising an experiment showing that only one of the two components of T2 actually enters the *E. coli* cell during infection **(Figure 16.4)**. In their experiment, they used a radioactive isotope of sulfur to tag protein in one batch of T2 and a radioactive isotope of phosphorus to tag DNA in a second batch. Because protein, but not DNA, contains sulfur, radioactive sulfur atoms were incorporated only into the protein of the phage. In a similar way, the atoms of radioactive phosphorus labeled only the DNA, not the protein, because nearly all the phage's phosphorus is in its DNA. In the experiment, separate samples of nonradioactive *E. coli* cells were infected separately with the protein-

labeled and DNA-labeled batches of T2. The researchers then tested the two samples shortly after the onset of infection to see which type of radioactively labeled molecule—protein or DNA—had entered the bacterial cells and would therefore be capable of reprogramming them.

Figure 16.4

Inquiry: Is protein or DNA the genetic material of phage T2?

Experiment

Alfred Hershey and Martha Chase used radioactive sulfur and phosphorus to trace the fates of protein and DNA, respectively, of T2 phages that infected bacterial cells. They wanted to see which of these molecules entered the cells and could reprogram them to make more phages.

Experiment. Is protein or D N A the genetic material of phage T 2?

Batch 1: Phages were grown with radioactive sulfur (superscript 35 S), which was incorporated into the phage protein (pink fluid inside a flask).

- 1. Inside flask, researchers mixed radioactively labeled phages with bacteria. The phages infected the bacterial cells. Bacterial cell drawn with phages (containing D N A) and radioactive proteins on the surface.
- 2. The mixture from the flask was agitated in a blender to free phage parts outside the bacteria from the cells. Drawing shows bacterial cell with phage D N A inside and empty protein shells and phages outside cell coming loose from bacterial cell.
- 3. Mixture centrifuged so bacteria (with phage D N A) formed a pellet at bottom of a test tube; free phages and page parts, which are lighter, remained suspended in the liquid.
- 4. Radioactivity measured in pellet and liquid; found in liquid (from phage protein).

Batch 2: Phages were grown with radioactive phosphorous (superscript 32 P), which was incorporated into phage D N A (blue parts in phages drawn and in liquid in a flask). Phages containing radioactive D N A (blue) shown in drawing on a bacterial cell.

- Mixture from flask agitated in blender, causing phages to come lose from outside bacterial cells, while bacterial cells contain phage D N A inside.
- Mixture from blender centrifuged; pellet at the bottom of a test tube contains bacteria and liquid contains free phages and phage parts, which are lighter.
- Radioactivity (phage D N A) found in pellet (because phage D N A is within bacterial cells).

Results. When proteins were labeled (batch 1, radioactivity remained outside the cells, but when D N A was labeled (batch 2, radioactivity was found inside the cells. Cells containing radioactive phage D N A released new phages with some radioactive phosphorus. Conclusion. Phage D N A entered bacterial cells, but phage proteins did not. Hershey and Chase concluded that D N A, not protein, functions as the genetic material of phage T 2.

Results

When proteins were labeled (batch 1), radioactivity remained outside the cells, but when DNA was labeled (batch 2), radioactivity was found inside the cells. Cells containing radioactive phage DNA released new phages with some radioactive phosphorus.

Conclusion

Phage DNA entered bacterial cells, but phage proteins did not. Hershey and Chase concluded that DNA, not protein, functions as the genetic material of phage T2.

Data from A. D. Hershey and M. Chase, Independent functions of viral protein and nucleic acid in growth of bacteriophage, *Journal of General Physiology* 36:39–56 (1952).

WHAT IF? *How would the results have differed if proteins carried the genetic information?*

*For suggested answer, see*Appendix A.**

Animation

The Hershey-Chase Experiment

image

Play

image

Hershey and Chase found that the phage DNA entered the host cells, but the phage protein did not. Moreover, when these bacteria were returned to a culture medium and the infection ran its course, the *E.*

coli released phages that contained some radioactive phosphorus. This result further showed that the DNA inside the cell played an ongoing role during the infection process. They concluded that the DNA injected by the phage must be the molecule carrying the genetic information that makes the cells produce new viral DNA and proteins. The Hershey-Chase experiment was a landmark study because it provided powerful evidence that nucleic acids, rather than proteins, are the hereditary material, at least for certain viruses.

Additional Evidence That DNA Is the Genetic Material

Further evidence that DNA is the genetic material came from the laboratory of biochemist Erwin Chargaff. DNA was known to be a polymer of nucleotides, each having three components: a nitrogenous (nitrogen-containing) base, a pentose sugar called deoxyribose, and a phosphate group **(Figure 16.5)**. The base can be adenine (A), thymine (T), guanine (G), or cytosine (C). Chargaff analyzed the base composition of DNA from a number of different organisms. In 1950, he reported that the base composition of DNA varies from one species to another. For example, he found that 32.8% of sea urchin DNA nucleotides have the base A, whereas only 24.7% of those from the bacterium *E. coli* have an A. This evidence of molecular diversity among species, which most scientists had presumed to be absent from DNA, made DNA a more credible candidate for the genetic material.

Figure 16.5

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The structure of a DNA strand.
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D N A strand from 5 prime end to 3 prime. From 5 prime end: backbone alternates phosphate group - sugar-phosphate group-sugar; a base (A, T, C, G) is attached to each sugar.

D N A consists of a phosphate-sugar backbone with bases (thymine, guanine, cytosine, or adenine) bonded to the sugars. Each repeating nucleotide monomer is made up of one phosphate group, one sugar, and one base. A phosphate group is a central P double bonded to one O and single bonded to 3 other Os. Two Os have negative charges. The third is bonded to the fifth C in deoxyribose (pentose sugar). Attached to carbon 1 of the sugar is the base (A, T, C or G). Attached to carbon 3 of the sugar is the next phosphate group in the backbone of the molecule. This structure repeats for the rest of the chain. The strand of D N A shown starts at the 5 prime end (with the first phosphate group) and ends at the 3 prime end (with the O H group of the last sugar). The base sequence shown here is T (single ring base), G (double ring base), C (single ring base), and A (double ring base).

Each DNA nucleotide monomer consists of the sugar deoxyribose (blue) attached to both a nitrogenous base (A, T, G, or C), and a phosphate group (yellow). The phosphate group of one nucleotide is attached to the

sugar of the next by a covalent bond, forming a “backbone” of alternating phosphates and sugars from which the bases project. A polynucleotide strand has directionality, from the 5′ end (with the phosphate group) to the 3′ end (with the —OH group of the sugar). 5′ and 3′ refer to the numbers assigned to the carbons in the sugar ring (see magenta numbers).

Animation

DNA and RNA Structure

image

Play

image

Chargaff also noticed a peculiar regularity in the ratios of nucleotide bases. In the DNA of each species he studied, the number of adenines approximately equaled the number of thymines, and the number of guanines approximately equaled the number of cytosines. In sea urchin DNA, for example, the four bases are present in these percentages: A=32.8% and T=32.1%; G=17.7% and C=17.3%. (The percentages are not exactly the same because of limitations in Chargaff’s techniques.)

These two findings became known as *Chargaff’s rules*: (1) DNA base composition varies between species, and (2) for each species, the percentages of A and T bases are roughly equal, as are those of G and C bases. In the **Scientific Skills Exercise**, you can use Chargaff’s rules to predict percentages of nucleotide bases. The basis for these rules remained unexplained until the discovery of the double helix.

Scientific Skills Exercise

Working with Data in a Table

AP Science Practices 1, 2, 4, 5, 6

Given the Percentage Composition of One Nucleotide in a Genome, Can We Predict the Percentages of the Other Three Nucleotides? Even before the structure of DNA was elucidated, Erwin Chargaff and his co-workers noticed a pattern in the base composition of nucleotides from different species: The percentage of adenine (A) bases roughly equaled that of thymine (T) bases, and the percentage of cytosine (C) bases roughly equaled that of guanine (G) bases. Further, the percentage of each pair (A–T or C–G) varied from species to species. We now know that the 1:1 A/T and C/G ratios are due to

complementary base pairing between A and T and between C and G in the DNA double helix, and differences between species are due to the unique sequences of bases along a DNA strand. In this exercise, you will apply Chargaff's rules to predict the composition of bases in a genome.

How the Experiments Were Done In Chargaff's experiments, DNA was extracted from the given species, hydrolyzed to break apart the nucleotides, and then analyzed chemically. These studies provided approximate values for each type of nucleotide. (Today, whole-genome sequencing allows base composition analysis to be done more precisely directly from the sequence data.)

Data from the Experiments Tables are useful for organizing sets of data representing a common set of values (here, percentages of A, G, C, and T) for a number of different samples (in this case, from different species). You can apply the patterns that you see in the known data to predict unknown values. In the table, complete base distribution data are given for sea urchin DNA and salmon DNA; you will use Chargaff's rules to fill in the rest of the table with predicted values.

Sea urchin.

Sea urchin

Predicting composition of nucleotides in genomes with the composition of one nucleotide.

Data from several papers by Chargaff: for example, E. Chargaff et al., Composition of the desoxypentose nucleic acids of four genera of sea-urchin, *Journal of Biological Chemistry* 195: 155–160 (1952).

Source of D N A	Base percentage
Adenine	Guanine
Sea urchin	32.8
Salmon	29.7
Wheat	28.1
E. coli	24.7
Human	30.4
Ox	29
Average percentage	blank

INTERPRET THE DATA

- Explain how the sea urchin and salmon data demonstrate both of Chargaff's rules.

- Using Chargaff's rules, fill in the table with your predictions of the missing percentages of bases, starting with the wheat genome and proceeding through *E. coli*, human, and ox. Show how you arrived at your answers.
- If Chargaff's rule—that the amount of A equals the amount of T and the amount of C equals the amount of G—is valid, then hypothetically we could extrapolate this to the combined DNA of all species on Earth (like one huge Earth genome). To see whether the data in the table support this hypothesis, calculate the average percentage for each base in your completed table by averaging the values in each column. Does Chargaff's equivalence rule still hold true?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

HHMI BioInteractive Animation

Chargaff's Ratio

image

Play

image

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

Building a Structural Model of DNA

Once most biologists were convinced that DNA was the genetic material, the challenge was to determine how the structure of DNA could account for its role in inheritance. By the early 1950s, the arrangement of covalent bonds in a nucleic acid polymer was well established (see Figure 16.5), and researchers focused on discovering the three-dimensional structure of DNA. Among the scientists working on the problem were Linus Pauling, at the California Institute of Technology, and Maurice Wilkins and Rosalind Franklin, at King's College in London. First to come up with the complete answer, however, were two scientists who were relatively unknown at the time—the American James Watson and the Englishman Francis Crick.

The collaborative work that solved the puzzle of DNA structure began soon after Watson went to Cambridge University, where Crick was studying protein structure with a technique called X-ray crystallography (see Figure 5.21). While visiting the laboratory of Maurice Wilkins, Watson saw an X-ray diffraction image of DNA produced by Wilkins's accomplished colleague Rosalind Franklin **(Figure 16.6)**. Images produced by X-ray crystallography are not actually pictures of molecules. The spots in the image were produced by X-rays that were diffracted (deflected) as they passed through aligned fibers of purified DNA. Watson was familiar

with the type of X-ray diffraction pattern of helical molecules, and seeing the photo that Wilkins showed him confirmed that DNA was helical in shape. The photo also added to earlier data obtained by Franklin and others suggesting the width of the helix and the spacing of the nitrogenous bases along it. The pattern in this photo implied that the helix was made up of two strands, contrary to a three-stranded model proposed by Linus Pauling a short time earlier. The presence of two strands accounts for the now-familiar term ****__double helix__. DNA is shown in some of its many different representations in **Figure 16.7**.

Figure 16.6

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Rosalind Franklin and her X-ray diffraction photo of DNA.
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Figure a is Rosalind Franklin. Figure b is Franklin’s X-ray diffraction of D N A. It has spaced segments in an X shape.

Advanced placement edition

Apply concepts and connect across Big Ideas using visual representations.

How does this model support common ancestry for all organisms? Given a distance of 0.34 nm between nucleotides, how long is the haploid DNA of a human with 3.2 billion nucleotides?

Figure 16.7

Visualizing DNA

D N A can be illustrated in many ways, but all diagrams represent the same basic structure. This figure compares structural images, simplified images, and D N A sequences.

These structural images show the three-dimensional shape of the D N A double helix (left) and chemical details of D N A's structure (right). Both images use the same colors for phosphate groups (yellow), deoxyribose sugars (blue), and nitrogenous bases (shades of green and orange).

Structural image on left: The D N A double helix is right-handed, as shown in this computer-generated space-filling model. Use your right hand (as shown in photo) to grasp the model and follow the sugar-phosphate backbone up the helix (red arrow) and around to the back. (It won't work with your left hand.) Phosphate groups are yellow, sugars are blue, and bases are dark orange, light orange, dark green and light green. Components tightly packed and twisted into double helix. Bases are 0.34 nanometers apart. There is one full turn every 10 base pairs (3.4 nanometers).

Structural image on right: Here, the two D N A strands are shown untwisted so it's easier to see the chemical details. Note that the strands are antiparallel, they are oriented in opposite directions, like the lanes of a divided street. Sugar-phosphate backbone on left runs (from top to bottom) 5 prime to three prime; backbone on the right runs from (top to bottom) 3 prime to 5 prime. Strands consist of repeating nucleotides (phosphate-sugar-base) with covalent bonds (solid lines) drawn between the phosphate group, sugar and base and hydrogen bonds (dashed lines) drawn between complementary bases (which form the rungs of the helix ladder.)

In this diagram, from the top left, is a phosphate group (5 prime end), pentose sugar, and T (single ring) hydrogen bonded (two dashed lines) to A (double ring), which is bonded to a pentose sugar (3 prime end), then a sugar-phosphate backbone.

Back to the left side: phosphate, sugar, G (double ring) hydrogen bonded (3 dashed lines) to a C (single ring), which is bonded to a sugar and then a phosphate group. Covalent sugar-phosphate bonds link the nucleotides of each strand.

Back to the left side: phosphate group-sugar-C (single ring) hydrogen bonded (3 dashed lines) to a G, then sugar-phosphate group. Hydrogen bonds (shown as dotted lines) between nitrogenous bases hold the strands together.

Back to the left side: phosphate group- sugar (3 prime end) bonded to an A (double ring) hydrogen bonded (2 dashed lines) to a T, then a sugar-phosphate group (5 prime end). Van der Waals interactions between stacked base pairs help hold the molecule together. When molecular detail is not necessary, D N A is portrayed in a range of simplified diagrams, depending on the focus of the figure. From left to right: ribbon models, ladder-style diagrams, and straight lines.

Ribbon models. The ribbons in these simplified double helix diagrams represent the sugar-phosphate backbones. Sugar-phosphate backbone shown as double strand of blue ribbon with bases on the inside. H-bonds drawn as dashed lined between complementary bases (A with T; and C with G).

Ladder-style diagrams. These flattened ladder style diagrams of D N A depict the sugar-phosphate backbones like the side rails of a ladder, with the base pairs as rungs. Light blue is used to indicate the more recently synthesized strand. Shown here are three ladder-style diagrams. In each diagram, each horizontal rung contains two base pairs: T with A, G with C, C with G, and A with T.

Straight lines. Sometimes the double-stranded D N A molecule is shown simply as two straight lines. Diagram shows two thin parallel vertical lines. The line on the left is labeled 5 prime at the top and 3 prime at the bottom. The line on the right is labeled 3 prime at the top and 5 prime at the bottom.

Genetic information is carried in D N A as a linear sequence of nucleotides that may be transcribed into m R N A and translated into a polypeptide. When focusing on the D N A sequence, each nucleotide can be

represented simply as the letter of its base: A, T, C, or G. In this example, starting from the left: 3 prime, A C G T A A G C G G T T A A T, 5 prime. On the next line, starting from the left: 5 prime, T G C A T T C G C C A A T T A, 3 prime.

Animation

DNA Double Helix

image

Play

image

Instructors: Additional questions related to this Visualizing Figure can be assigned in **Mastering Biology**.

Watson and Crick began building models of a double helix that would conform to the X-ray measurements and what was then known about the chemistry of DNA, including Chargaff's rule of base equivalences. Having also read an unpublished annual report summarizing Franklin's work, they knew she had concluded that the sugar-phosphate backbones were on the outside of the DNA molecule, contrary to their working model. Franklin's arrangement was appealing because it put the negatively charged phosphate groups facing the aqueous surroundings, while the relatively hydrophobic nitrogenous bases were hidden in the interior. Watson constructed such a model, in which the two sugar-phosphate backbones are ******antiparallel**—that is, their subunits run in opposite directions (see Figure 16.7). You can imagine the overall arrangement as a rope ladder with rigid rungs. The side ropes represent the sugar-phosphate backbones, and the rungs represent pairs of nitrogenous bases. Now imagine twisting the ladder to form a helix. Franklin's X-ray data indicated that the helix makes one full turn every 3.4 nm along its length. With the bases stacked just 0.34 nm apart, there are ten layers of base pairs, or rungs of the ladder, in each full turn of the helix.

The nitrogenous bases of the double helix are paired in specific combinations: adenine (A) with thymine (T), and guanine (G) with cytosine (C). It was mainly by trial and error that Watson and Crick arrived at this key feature of DNA. At first, Watson imagined that the bases paired like with like—for example, A with A and C with C. But this model did not fit the X-ray data, which suggested that the double helix had a uniform diameter. Why is this requirement inconsistent with like-with-like pairing of bases? Adenine and guanine are purines, nitrogenous bases with two organic rings, while cytosine and thymine are nitrogenous bases called pyrimidines, which have a single ring. Pairing a purine with a pyrimidine is the only combination that results in a uniform diameter for the double helix **(Figure 16.8)**.

Figure 16.8

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Possible base pairings in the DNA double helix.
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Dashed red lines mark width of D N A from X-ray data. Purine + pyrimidine (double ring + single ring) is the only combination of bases that is consistent with X-ray data.

Dashed red lines show the width of D N A from X-ray data. Purine + purine = double ring + double ring = too wide (wider than dashed lines). Pyrimidine + pyrimidine = single ring + single ring = too narrow (narrower than dashed lines). Purine + pyrimidine = fits dashed lines and is consistent with X-ray data.

Watson and Crick reasoned that there must be additional specificity of pairing dictated by the structure of the bases. Each base has chemical side groups that can form hydrogen bonds with its appropriate partner: Adenine can form two hydrogen bonds with thymine and only thymine; guanine forms three hydrogen bonds with cytosine and only cytosine. In shorthand, A pairs with T, and G pairs with C **(Figure 16.9)**.

Figure 16.9

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Base pairing in DNA.
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Two black dotted lines (hydrogen bonds) form between Adenine and Thymine, and three black dotted lines (hydrogen bonds) form between Guanine and Cytosine.

Hydrogen bonds form between chemical side groups of complementary nitrogenous base pairs in the D N A molecule. Between A and T, hydrogen bonds form between an H on A and an O on T, and between an N on A and an H on T. Two total hydrogen bonds form between A and T. Between G and C, hydrogen bonds from between an O on G and an H on C, and between an H on G and an N on C, and between an H on G and an O on C. Three total hydrogen bonds form between G and C.

The Watson-Crick model took into account Chargaff's ratios and ultimately explained them. Wherever one strand of a DNA molecule has an A, the partner strand has a T. Similarly, a G in one strand is always paired with a C in the complementary strand. Therefore, in the DNA of any organism, the amount of adenine equals the amount of thymine, and the amount of guanine equals the amount of cytosine. (Modern DNA-sequencing techniques have confirmed that the amounts are exactly equal.) Although the base-pairing rules dictate the combinations of nitrogenous bases that form the "rungs" of the double helix, they do not restrict the sequence of nucleotides *along* each DNA strand. The linear sequence of the four bases can be varied in countless ways, and each gene has a unique base sequence.

In April 1953, Watson and Crick surprised the scientific world with a succinct, one-page paper that reported their molecular model for DNA: the double helix, which has since become an icon of molecular biology.

Watson and Crick, along with Maurice Wilkins, were awarded the Nobel Prize in 1962 for this work. (Sadly, Rosalind Franklin had died at the age of 37 in 1958 and was thus ineligible for the prize.) The beauty of the double helix model was that the structure of DNA suggested the basic mechanism of its replication.

HHMI BioInteractive Video

Great Discoveries in Science: The Double Helix

image

image

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Concept 16.2: Many proteins work together in DNA replication and repair

Hereditary information in DNA directs the development of your biochemical, anatomical, physiological, and, to some extent, behavioral traits. Your resemblance to your parents has its basis in the accurate replication of DNA prior to meiosis and therefore its transmission from your parents' generation to yours. Replication prior to mitosis ensures faithful transmission of genetic information from a parent cell to two daughter cells.

Of all nature's molecules, nucleic acids are unique in their ability to dictate their own replication from monomers. The relationship between structure and function is evident in the double helix: The specific complementary pairing of nitrogenous bases in DNA has a functional significance. Watson and Crick ended their classic paper with this statement: "It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material."1 In this section, you will learn about the basic principle of **DNA replication**, the copying of DNA, as well as some important details of the process.

The Basic Principle: Base Pairing to a Template Strand

In a second paper, Watson and Crick stated their hypothesis for how DNA replicates: "Now our model for deoxyribonucleic acid is, in effect, a pair of templates, each of which is complementary to the other. We imagine that prior to duplication the hydrogen bonds are broken, and the two chains unwind and separate.

Each chain then acts as a template for the formation on to itself of a new companion chain, so that eventually we shall have two pairs of chains, where we only had one before. Moreover, the sequence of the pairs of bases will have been duplicated exactly."2

Figure 16.10 illustrates the basic idea. If you cover a DNA strand in Figure 16.10a, its linear sequence of nucleotides is revealed by applying the base-pairing rules to the uncovered strand. The two strands are complementary; each stores the information necessary to reconstruct the other. When a cell copies a DNA molecule, each strand serves as a template for ordering nucleotides into a new, complementary strand. Nucleotides line up along the template strand according to the base-pairing rules and are linked to form the new strands. One double-stranded DNA molecule becomes two, each an exact replica of the "parental" molecule.

Figure 16.10

A model for DNA replication: the basic concept.

A model for D N A replication.

In this simplified illustration, a short segment of DNA has been untwisted. Simple shapes symbolize the four kinds of bases. Dark blue represents DNA strands present in the parental molecule; light blue represents newly synthesized DNA.

The parts are as follows.

Part a. The parental molecule has two complementary strands of D N A. Each base is paired by hydrogen bonding with its specific partner, A with T and G with C. D N A shown resembling ladder with bases as shapes in rungs. From top, base pairs: A T, C G, T A, A T, and G C. A and G are larger shapes than C and T.

Part b. First, the two D N A strands are separated. Each parental strand can now serve as a template for a new, complementary strand. Strands separated so that left half reads (top to bottom): A C T A G and right half reads (top to bottom) T G A T C.

Part c. Nucleotides complementary to the parental (dark blue) strand are connected to form the sugar-phosphate backbones of the new daughter (light blue) strands. Two new strands of D N A formed; each is half parental strand and half new strand made of new nucleotide bases. First D N A molecule: Parental strand is A C T A G. New strand is T G A T C. Complementary base pairs are back together. Second D N A Molecule: Parental strand is T G A T C. New strand is A C T A G. Complementary base pairs are back together.

Animation

DNA Replication: An Overview

image

image

This model of DNA replication remained untested for several years following publication of the DNA structure. The necessary experiments were simple in concept but difficult to perform. Watson and Crick's model predicts that when a double helix replicates, each of the two daughter molecules will have one old strand, from the parental molecule, and one newly made strand. This **semiconservative model** can be distinguished from a *conservative model* of replication, in which the two parental strands somehow come back together after the process (that is, the parental molecule is conserved). In yet a third model, called the *dispersive model*, all four strands of DNA following replication have a mixture of old and new DNA **(Figure 16.11)**.

Figure 16.11

DNA replication: three alternative models.

D N A replication: 3 alternative models.

Each short segment of double helix symbolizes the DNA within a cell. Beginning with a parent cell, we follow the DNA for two more generations of cells—two rounds of DNA replication. Parental DNA is dark blue; newly made DNA is light blue.

The parts are as follows.

Part a. Conservative model. The two parental strands reassociate after acting as templates for new strands, thus restoring the parental double helix. Start: Parental D N A molecule (shown as a ribbon double helix); follow for two replications. Result is one parental D N A molecule and 3 new molecules. Parental D N A molecule undergoes first replication; result is a parental D N A molecule and a new D N A molecule. Each molecule then goes through a second replication, resulting in two new D N A molecules from the new D N A molecule, and a new molecule and a parental molecule from the parental molecule.

Part b. Semiconservative model. The two strands of the parental molecule separate, and each functions as a template for synthesis of a new, complementary strand. Start: parental D N A molecule; follow for two replications. Result is two D N A molecules that are half parental strand D N A and half new strand, and two D N A molecules that are new. Parental D N A molecule undergoes first replication; result is two D N A molecules that are half parental strand D N A and half new strand D N A. Each of these molecules then undergoes a second replication; result is, for each, one D N A molecule that is half parental strand and half new strand D N A and one D N A molecule that is new.

Part c. Dispersive model. Each strand of both daughter molecules contains a mixture of old and newly synthesized D N A. Start: Parental D N A molecule; follow for two replications. Result is four D N A molecules that contain both parental and new D N A. Parental D N A undergoes first replication; result is two D N A molecules that contain both parental and new D N A. These molecules then undergo a second replication; result is (for each) 2 D N A molecules that contain both parental and new D N A.

Although mechanisms for conservative or dispersive DNA replication are not easy to devise, these models remained possibilities until they could be ruled out. After two years of preliminary work at the California Institute of Technology in the late 1950s, Matthew Meselson and Franklin Stahl devised a clever experiment that distinguished between the three models, described in **Figure 16.12**. Their results supported the semiconservative model of DNA replication, as predicted by Watson and Crick, and their experiment is widely recognized among biologists as a classic example of elegant design.

Figure 16.12

Inquiry: Does DNA replication follow the conservative, semiconservative, or dispersive model?

Experiment

Matthew Meselson and Franklin Stahl cultured *E. coli* for several generations in a medium containing nucleotide precursors labeled with a heavy isotope of nitrogen, 15N. They then transferred the bacteria to a medium with only 14N, a lighter isotope. They took one sample after the first DNA replication and another after the second replication. They extracted DNA from the bacteria in the samples and then centrifuged each DNA sample to separate DNA of different densities.

Results

A diagram of the experiment.

Flasks show bacteria cultured in medium 15 N, a heavy isotope, and then transferred to another flask with medium 14 N, a lighter isotope. Test tubes from the lighter isotope medium are centrifuged. After the second replication the centrifuged test tube has contents that are more dense at the bottom and less dense on top.

Conclusion

Meselson and Stahl compared their results to those predicted by each of the three models in Figure 16.11, as summarized in the following diagram. The first replication in the 14N medium produced a band of many molecules of hybrid 15N-14N DNA. This result eliminated the conservative model. The second replication produced both light and hybrid DNA, refuting the dispersive model and supporting the semiconservative model. They therefore concluded that DNA replication is semiconservative.

A diagram of experiment predictions.

The predictions are as follows.

Conservative Model: Test tube with band of hybrid D N A eliminates conservative model. Predictions would be: First replication: one parental D N A and one new D N A molecule (test tube with light band high and dark band low); 2nd replication: One parental and 3 new D N A molecules (test tube with light band high and dark band low).

Semiconservative model: Test tube with band of hybrid D N A supports semiconservative model. Predictions would be: First replication: Two molecules of D N A that are half parental and half new (test tube with band in the middle). Second replication: Two molecules of D N A that are half hybrid and half new and two molecules that are new (test tube with light band high and band in the middle).

Dispersive model: Band with all hybrid D N A following the second replication eliminates the dispersive model.

Predictions would be: First replication: two molecules of D N A that are both parental and new (test tube with band in the middle). Second replication: all D N A molecules are part parental and part new (test tube with band up high).

Data from M. Meselson and F. W. Stahl, The replication of DNA in *Escherichia coli, Proceedings of the National Academy of Sciences USA* 44:671–682 (1958).

INQUIRY IN ACTION Read and analyze the original paper in*Inquiry in Action: Interpreting Scientific Papers*.

Instructors: A related Experimental Inquiry Tutorial can be assigned in **Mastering Biology**.

WHAT IF? *If Meselson and Stahl had first grown the cells in*N14*-containing medium and then moved them into*N15*-containing medium before taking samples, what would have been the result after each of the two replications?*

*For suggested answer, see**Appendix A*.

The basic principle of DNA replication is conceptually simple. However, the actual process involves some complicated biochemical gymnastics, as we will now see.

DNA Replication: *A Closer Look*

The bacterium *E. coli* has a single chromosome of about 4.6 million nucleotide pairs. In a favorable environment, an *E. coli* cell can copy all of this DNA and divide to form two genetically identical daughter cells in considerably less than an hour. Each of *your* somatic cells has 46 DNA molecules in its nucleus, one long double-helical molecule per chromosome. In all, that represents about 6 billion nucleotide pairs, or over 1,000 times more DNA than is found in most bacterial cells. If we were to print the one-letter symbols for these bases (A, G, C, and T) the size of the type you are now reading, the 6 billion nucleotide pairs of information in a diploid human cell would fill about 1,400 biology textbooks. Yet it takes one of your cells just a few hours to copy all of this DNA during S phase of interphase. This replication of an enormous amount of genetic information is achieved with very few errors—only about one per 10 billion nucleotides. The copying of DNA is remarkable in its speed and accuracy.

More than a dozen enzymes and other proteins participate in DNA replication. Much more is known about how this "replication machine" works in bacteria (such as *E. coli*) than in eukaryotes, and we will describe the basic steps of the process for *E. coli*, except where otherwise noted. What scientists have learned about eukaryotic DNA replication suggests, however, that most of the process is fundamentally similar for prokaryotes and eukaryotes.

Getting Started

The replication of chromosomal DNA begins at particular sites called **origins of replication**, short stretches of DNA that have a specific sequence of nucleotides. The *E. coli* chromosome, like many other bacterial chromosomes, is circular and has a single origin. Proteins that initiate DNA replication recognize this sequence and attach to the DNA, separating the two strands and opening up a replication "bubble" **(Figure 16.13a)**. Replication of DNA then proceeds in both directions until the entire molecule is copied. In contrast to a bacterial chromosome, a eukaryotic chromosome may have hundreds or even a few thousand replication origins. Multiple replication bubbles form and eventually fuse, thus speeding up the copying of the very long DNA molecules **(Figure 16.13b)**. As in bacteria, eukaryotic DNA replication proceeds in both directions from each origin.

Figure 16.13

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Origins of replication in *E. coli* and eukaryotes.
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Origins of replication in E. coli and eukaryotes.

The red arrows indicate the movement of the replication forks and thus the overall directions of DNA replication within each bubble.

Part a. Origin of replication in an E. coli cell. The circular chromosome of E. coli and other bacteria have only one origin of replication. The parental strands separate there, forming a replication bubble with two forks (red arrows). Replication proceeds in both directions until the forks meet on the other side, resulting in two daughter D N A molecules. The T E M micrograph shows a bacterial chromosome with a replication bubble. T E M Scale: 0.5 micrometers.

Part b. Origins of replication in a eukaryotic cell. In a linear chromosome of a eukaryote, replication bubbles form at many sites along the giant D N A molecule during S phase of interphase. The bubbles expand as replication proceeds in both directions (red arrows). Eventually, the bubbles fuse and synthesis of the daughter strands is complete. The T E M micrograph shows three replication bubbles along the D N A of a cultured Chinese hamster cell. T E M scale: 0.25 micrometers.

DRAW IT*In the TEM, add arrows in the forks of the third bubble.*

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The Replication Fork in *E. coli*

image

image

At each end of a replication bubble is a **replication fork**, a Y-shaped region where the parental strands of DNA are being unwound. Several kinds of proteins participate in the unwinding **(Figure 16.14)**. **Helicases** are enzymes that untwist the double helix at the replication forks, separating the two parental strands and making them available as template strands. After the parental strands separate, **single-strand binding proteins** bind to the unpaired DNA strands, keeping them from re-pairing. The untwisting of the double helix causes tighter twisting and strain ahead of the replication fork. **Topoisomerase** is an enzyme that helps relieve this strain by breaking, swiveling, and rejoining DNA strands.

Figure 16.14

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    Some of the proteins involved in the initiation of DNA
replication.
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Activities of the proteins primase, single-strand binding proteins, helicase and topoisomerase are illustrated.

The same proteins function at both replication forks in a replication bubble. For simplicity, only the left-hand fork is shown, and the DNA bases are drawn much larger in relation to the proteins than they are in reality.

D N A molecule is drawn open at one end to illustrate the inside of a replication bubble. Starting at the 3 prime end of D N A, Primase synthesizes R N A primers (from 5 prime to 3 prime ends) using the parental D N A as a template. Single-strand binding proteins are drawn inside replication bubble, stabilizing unwound parental strands. Helicase (shown at the replication fork) unwinds and separates the parental D N A strands. Topoisomerase (drawn on the wound D N A molecule ahead of the replication bubble) breaks, swivels, and rejoins the parental D N A ahead of the replication fork, relieving the strain caused by unwinding.

Synthesizing a New DNA Strand

The unwound sections of parental DNA strands are now available to serve as templates for the synthesis of new complementary DNA strands. However, the enzymes that synthesize DNA cannot *initiate* the synthesis of a polynucleotide; they can only add DNA nucleotides to the end of an already existing chain that is base-paired with the template strand. An initial nucleotide chain that can be used as a pre-existing chain is produced during DNA synthesis; this is actually a short stretch of RNA, not DNA. The RNA chain is called a **primer** and is synthesized by the enzyme **primase** (see Figure 16.14). Primase starts a complementary RNA chain with a single RNA nucleotide and adds RNA nucleotides one at a time, using the parental DNA strand as a template. The completed primer, generally five to ten nucleotides long, is thus base-paired to the template strand. The new DNA strand will start from the 3′ end of the RNA primer.

Enzymes called **DNA polymerases** catalyze the synthesis of new DNA by adding nucleotides to the 3′ end of a preexisting chain. In *E. coli*, there are several DNA polymerases, but two of them appear to play the major roles in DNA replication: DNA polymerase III and DNA polymerase I. The situation in eukaryotes is more complicated, with at least 11 different DNA polymerases discovered so far, although the general principles are the same.

Most DNA polymerases require a primer and a DNA template strand, along which complementary DNA nucleotides are lined up, one by one. In *E. coli*, DNA polymerase III (abbreviated DNA pol III) adds a DNA nucleotide to the RNA primer and then continues adding DNA nucleotides, which are complementary to the parental DNA template strand, to the growing end of the new DNA strand. The rate of elongation is about 500 nucleotides per second in bacteria and 50 per second in human cells.

Each nucleotide to be added to a growing DNA strand consists of a sugar attached to a base and to three phosphate groups. You have already encountered such a molecule—ATP (adenosine triphosphate; see Figure 8.9). The only difference between the ATP of energy metabolism and dATP, the adenine nucleotide used to make DNA, is the sugar component, which is deoxyribose in the building block of DNA but ribose in ATP.

Like ATP, the nucleotides used for DNA synthesis are chemically reactive, partly because their triphosphate tails have an unstable cluster of negative charge. DNA polymerase catalyzes the addition of each monomer to the growing end of a DNA strand by a condensation reaction in which two phosphate groups are lost as a molecule of pyrophosphate. Subsequent hydrolysis of the pyrophosphate to two molecules of inorganic phosphate is a coupled exergonic reaction that helps drive the polymerization reaction **(Figure 16.15)**.

Figure 16.15

Addition of a nucleotide to a DNA strand.

D N A template drawn from (top to bottom) 3 prime to 5 prime. New strand drawn from 5 prime to 3 prime. Nucleotide is added to 3 prime end of new strand.

D N A template drawn from (top to bottom) 3 prime to 5 prime. New strand drawn from 5 prime to 3 prime. Nucleotide is added to 3 prime end of new strand (on the bottom in this diagram). Both template and new strands have sugar-phosphate backbones, with the sugars attached to bases. From top to bottom, base pairs shown are (new strand on left, template strand on right): A-T, C-G, G-C, followed by an A on the template strand. A nucleotide with the base T is about to be added to the new strand at the 3 prime end. The nucleotide being added consists of a T attached to deoxyribose sugar, which is attached to 3 phosphate groups. D N A polymerase adds this nucleotide to the growing new strand at the 3 prime end, so that the T can pair with A in the template strand, and two phosphate groups are released in the process, forming pyrophosphate. Arrows show pyrophosphate going to 2 inorganic phosphates.

DNA polymerase catalyzes addition of a nucleotide to the 3′ end of a growing DNA strand, with the release of two phosphates.

DRAW IT*Circle the area where the new bond was made.*

*For suggested answer, see*Appendix A.**

Figure 16.15 Walkthrough

Addition of a nucleotide to a DNA strand

Watch an explanation of this figure in the Figure 16.15 Walkthrough: Addition of a nucleotide to a DNA strand.

image

image

Antiparallel Elongation

As we have noted previously, the two ends of a DNA strand are different, giving each strand directionality, like a one-way street (see Figure 16.5). In addition, the two strands of DNA in a double helix are antiparallel, meaning that they are oriented in opposite directions to each other, like the two sides of a divided street (see Figure 16.15). Therefore, the two new strands formed during DNA replication must also be antiparallel to their template strands.

The antiparallel arrangement of the double helix, together with a constraint on the function of DNA polymerases, has an important effect on how replication occurs. Because of their structure, DNA polymerases can add nucleotides only to the free 3′ end of a primer or growing DNA strand, never to the 5′ end (see Figure 16.15). Thus, a new DNA strand can elongate only in the 5′→3′ direction. With this in mind, let's examine one of the two replication forks in a bubble **(Figure 16.16)**. Along one template strand, DNA polymerase III can synthesize a complementary strand continuously by elongating the new DNA in the mandatory 5′→3′ direction. DNA pol III remains in the replication fork on that template strand and continuously adds nucleotides to the new complementary strand as the fork progresses. The DNA strand made by this mechanism is called the **leading strand**. Only one primer is required for DNA pol III to synthesize the entire leading strand (see Figure 16.16).

Figure 16.16

Synthesis of the leading strand during DNA replication.

Overview: bubble with two replication forks; replication proceeds in both directions; leading and lagging strands labeled. Zoomed in detail: synthesis of the leading strand.

This diagram focuses on the left replication fork shown in the overview box. DNA polymerase III (DNA pol III), shaped like a cupped hand, is shown closely associated with a protein called the "sliding clamp" that encircles the newly synthesized double helix like a doughnut. The sliding clamp moves DNA pol III along the DNA template strand.

Arrows show leading strands synthesized continuously (solid line) from the origin toward the replication forks in one direction, and lagging strands synthesized towards the origin away from the replication forks in pieces (dashed lines). Small sections on all new strands of D N A are labeled as primer at the start of replication.

1. Arrows zoom in from leading strand in the overview to parental D N A, showing that after R N A primer is made, D N A pol 3 starts to synthesize the leading strand. Origin of replication is at the 3 prime end of the parental D N A; R N A primer is made from this template from 5 prime to 3 prime. The sliding clamp encircles both the D N A and the R N A; just beyond the sliding clamp

(towards the 3 prime end of the new molecule) is D N A pol 3. Single-stranded binding proteins ahead of D N A pol 3 stabilize the unwound parental strands near the replication fork. On the 5 prime end of the parental D N A, single stranded binding proteins and helicase are shown.

2. The leading strand is elongated continuously in the 5 prime to 3 prime direction as the fork progresses. R N A primer shown is only a few bases long; after the primer is new D N A. The sliding clamp is behind D N A pol 3, moving towards the replication fork and the 3 prime end of the new D N A strand. Single stranded binding proteins stabilize the parental strands; helicase unwinds and separates parental D N A at the fork.

BioFlix® Animation

Synthesis of the Leading Strand

image

image

To elongate the other new strand of DNA in the mandatory 5′→3′ direction, DNA pol III must work along the other template strand in the direction *away from* the replication fork. The DNA strand elongating in this direction is called the **lagging strand**. In contrast to the leading strand, which elongates continuously, the lagging strand is synthesized discontinuously, as a series of segments. These segments of the lagging strand are called **Okazaki fragments**, after Reiji Okazaki, the Japanese scientist who discovered them. The fragments are about 1,000–2,000 nucleotides long in *E. coli* and 100–200 nucleotides long in eukaryotes.

Figure 16.17 illustrates the steps in the synthesis of the lagging strand at one fork. Whereas only one primer is required on the leading strand, each Okazaki fragment on the lagging strand must be primed separately (**STEPS 1** and **4**). After DNA pol III forms an Okazaki fragment (**STEPS 2** to **4**), another DNA polymerase, DNA pol I, replaces the RNA nucleotides of the adjacent primer with DNA nucleotides one at a time (**STEP 5**). But DNA pol I cannot join the final nucleotide of this replacement DNA segment to the first DNA nucleotide of the adjacent Okazaki fragment. Another enzyme, **DNA ligase**, accomplishes this task, joining the sugar-phosphate backbones of all the Okazaki fragments into a continuous DNA strand (**STEP 6**).

Figure 16.17

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Synthesis of the lagging strand.
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Overview: bubble with two replication forks; replication proceeds in both directions; leading and lagging strands labeled. Zoomed in detail: synthesis of lagging strands.

Overview: Replication bubble with two replication forks. Origin of replication labeled in the center of the bubble on top and bottom parental strands. Arrows show leading strands synthesized continuously (solid line) from the origin toward the replication forks in one direction, and lagging strands synthesized towards the origin away from the replication forks in pieces (dashed lines). Small sections on all new strands of D N A are primer at the start of replication.

- 1. Primase joins RNA nucleotides into primer. Parental D N A strand drawn with 3 prime (left side) and 5 prime (right side) ends labeled. Primase encircles parental D N A and builds RNA primer from 5 prime to 3 prime, moving towards the origin of replication. There are several parental D N A bases in between the primer and the origin.

- 2. RNA primer from step 1 drawn on parental template; only 5 bases long in this illustration. After the primer new D N A is shown on the parental template, labeled as the 1st Okazaki fragment. New D N A is built from 5 prime to 3 prime. On the left is the sliding clamp; adjacent to the sliding clamp is D N A pol 3 (adding nucleotides to form Okazaki fragment 1. Towards the 5 prime end of the D N A (to the right) another RNA primer fragment is drawn.

- 3. Okazaki fragment 1 has reached the next primer to the right (towards the 5 prime end of the parental D N A molecule) and D N A pol 3 detaches.

- 4. Parental D N A template shown (left to right) 3 prime to 5 prime. RNA primer for fragment 2 is built from 5 prime to 3 prime (left to right) starting at the 3 prime end (left) of the parental D N A template. After a few base length of RNA primer, Okazaki fragment 2 begins. The sliding clamp is left of D N A pol 3, which adds D N A nucleotides to the growing fragment 2. D N A pol 3 will detach when it reaches the fragment 1 primer (to the right, towards the 5 prime end of the parental D N A template).

- 5. D N A parental strand runs (left to right) 3 prime to 5 prime. New strand runs along the template from 3 prime (left) to 5 prime (right). D N A pol 1 replaces the RNA primer from the fragment 1 primer (in the middle of the molecule) with D N A nucleotides.

- 6. D N A ligase forms a bond between the newest D N A and the D N A of fragment 1.

- 7. The lagging strand in this region is now complete. Overall direction of replication is toward the left, but lagging strand was synthesized toward the right. Fragments 1 and 2 are shown together on the parental D N A template strand. The parental D N A template strand runs (left to right) 3 prime to 5 prime, while the new strand runs (left to right) 5 prime to 3 prime. The new strand consists of, from left to right, RNA primer for a few base lengths, then fragment 2 joined to fragment 1.

BioFlix® Animation

Synthesis of the Lagging Strand

image

image

Synthesis of the leading strand and synthesis of the lagging strand occur concurrently and at the same rate. The lagging strand is so named because its synthesis is delayed slightly relative to synthesis of the leading strand; each new fragment of the lagging strand cannot be started until enough template has been exposed at the replication fork.

CheckPoint

Leading and Lagging Strand Synthesis

Check your understanding before you move on, with the CheckPoint: Leading and Lagging Strand Synthesis.

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Figure 16.18 and **Table 16.1** summarize DNA replication. Please study them carefully before proceeding.

Figure 16.18

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A summary of bacterial DNA replication.
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Overview: bubble with two replication forks; replication proceeds in both directions. Zoomed in detail: synthesis of the leading and the lagging strands.

The detailed diagram shows the left-hand replication fork of the replication bubble shown in the overview (upper right). Viewing each daughter strand in its entirety in the overview, you can see that half of it is

made continuously as the leading strand, while the other half (on the other side of the origin) is synthesized in fragments as the lagging strand.

Overview: Replication bubble with two replication forks. Origin of replication labeled in the center of the bubble on top and bottom parental strands. Arrows show leading strands synthesized continuously (solid line) from the origin toward the replication forks in one direction, and lagging strands synthesized towards the origin away from the replication forks in pieces (dashed lines). Small sections on all new strands of D N A are primer at the start of replication.

- 1. At the left replication fork, helicase unwinds the parental double helix.
- 2. Near the left replication fork, molecules of single-strand binding protein stabilize the unwound template strands.
- 3. The parental D N A template is shown unwound near the top of the replication bubble. Synthesis of the leading strand from this template begins at the origin of replication (top middle of bubble) and moves towards the left replication fork, from the 5 prime to 3 prime end of the new D N A molecule. Near the left replication fork, the sliding clamp moves D N A pol 3 along the template, adding D N A nucleotides to the 3 prime end of the new D N A and towards the left replication fork.
- 4. On the left is the replication fork, and the 3 prime end of the parental D N A template. Helicase is drawn at the fork. Just to the right of the fork is a single strand binding protein and then a short section of R N A primer, which primase builds from the 5 prime (left) to 3 prime (right) end on the parental D N A template. This new strand will be the fifth Okazaki fragment.
- 5. When D N A pol 3 reaches the R N A primer on fragment 3, it will detach and begin adding D N A nucleotides to the 3 prime end of the fragment 5 primer in the replication fork. Primer is being synthesized on the bottom strand of the replication fork, with the sliding clamp pushing D N A pol 3 along the template, adding nucleotides to the 3 prime end of the new strand. In the space of a few template bases, D N A pol 3 will reach the 5 prime end of the primer of lagging strand number 3 to the right. Fragment 5 is to the left, towards the replication fork.
- 6. D N A pol 1 removes primer from 5 prime end of fragment 2, replacing it with D N A nucleotides added one by one to the 3 prime end of fragment 3. After the last addition,

the backbone of new D N A fragment 1 is left with a free 3 prime end.

- 7. D N A ligase joins the 3 prime end of fragment 2 to the 5 prime end of fragment 1. At this point, the bottom of the replication bubble appears: Parental D N A template runs (left to right) 3 prime to 5 prime. Helicase is at the left replication fork, followed to the right by a single strand binding protein. Just to the right, primase adds R N A primer from 5 prime to 3 prime in what will become the 5th Okazaki fragment. To the right, after space of parental template D N A nucleotides only, a section of R N A primer is shown from 5 prime towards the 3 prime end, and a sliding clamp pushes D N A pol 3 along the template, adding D N A nucleotides at the 3 prime end of the new strand. To the right, after a space of a few parental D N A template nucleotides, is another short section of R N A primer is followed by lagging strand fragment 3, and then to the right D N A pol 1 which is removing primer from fragment 2 and adding D N A nucleotides. To the right of fragment 2 is fragment 1, which is being joined to fragment 2 by D N A ligase. The 3 prime end of the backbone of fragment 1 will remain free.

DRAW IT*Draw a diagram showing the right-hand fork of the bubble in this figure. Number the Okazaki fragments, and label all*5′*and*3′*ends.*

*For suggested answer, see*Appendix A.*

Animation

DNA Replication: A Closer Look

image

Table 16.1

Bacterial DNA Replication Proteins and Their Functions

A table titled Table 16.1 Bacterial D N A replication proteins and their functions illustrates various proteins and their functions.

Table 16.1 Bacterial D N A Replication Proteins and their Functions.
Protein
Helicase

Single-stranded binding protein
Topoisomerase
Primase
D N A polymerase Roman numeral 3
D N A polymerase Roman numeral 1
D N A ligase

The DNA Replication Complex

It is traditional—and convenient—to represent DNA polymerase molecules as locomotives moving along a DNA railroad track, but such a model is inaccurate in two important ways. First, the various proteins that participate in DNA replication actually form a single large complex, a “DNA replication machine.” Many protein-protein interactions facilitate the efficiency of this complex. For example, by interacting with other proteins at the fork, primase apparently acts as a molecular brake, slowing progress of the replication fork and coordinating the placement of primers and the rates of replication on the leading and lagging strands. Second, the DNA replication complex may not move along the DNA; rather, the DNA may move through the complex during the replication process. In eukaryotic cells, multiple copies of the complex, perhaps grouped into “factories,” may be anchored to the nuclear matrix, a framework of fibers extending through the interior of the nucleus.

Some experimental evidence supports a model in which two DNA polymerase molecules, one on each template strand, “reel in” the parental DNA and extrude newly made daughter DNA molecules. In this so-called *trombone model*, the lagging strand is also looped back through the complex **(Figure 16.19)**. Whether the complex moves along the DNA or whether the DNA moves through the complex, either anchored or not, are still open, unresolved questions that are under active investigation.

Figure 16.19

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The “trombone” model of the DNA replication complex.
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D N A molecule with section unwound for replication. D N A pol 3, helicase, single strand binding proteins, connecting proteins, and D N A pol 1 all shown.

In this proposed model, two molecules of DNA polymerase III work together in a complex, one on each strand, with helicase and other proteins. The lagging strand template DNA loops through the complex,

resembling the slide of a trombone.

At the replication fork top parental template strand runs 5 prime to 3 prime (left to right) and bottom parental template strands drawn 3 prime to 5 prime. The leading strand is synthesized from the top parental D N A template strand in the 5 prime to 3 prime direction (towards the replication fork). D N A pol 3 adds nucleotides to the 3 prime end of the leading strand. Connecting proteins hold strands of parental template D N A in place. Single strand binding proteins stabilize the template strands. Helicase unwinds parental D N A at the replication fork. The lagging strand is synthesized from the bottom parental template strand. Primase synthesizes R N A primer from 5 prime to 3 prime. Another D N A pol 3 adds nucleotides to the lagging strand fragments. D N A pol 1 removes R N A nucleotides of primer from 5 prime end and replaces them with D N A nucleotides.

DRAW IT *Draw a line tracing the lagging strand template in this figure.*

*For suggested answer, see*Appendix A.**

BioFlix® Animation

DNA Replication

image

Proofreading and Repairing DNA

We cannot attribute the accuracy of DNA replication solely to the specificity of base pairing. Initial pairing errors between incoming nucleotides and those in the template strand occur at a rate of one in 105 nucleotides. However, errors in the completed DNA molecule amount to only one in 1010 10 billion nucleotides, an error rate that is 100,000 times lower. This is because during DNA replication, many DNA polymerases proofread each nucleotide against its template as soon as it is covalently bonded to the growing strand. Upon finding an incorrectly paired nucleotide, the polymerase removes the nucleotide and then resumes synthesis. (This action is similar to fixing a texting error by deleting the wrong letter and then entering the correct one.)

Mismatched nucleotides sometimes evade proofreading by a DNA polymerase. In **mismatch repair**, other enzymes remove and replace incorrectly paired nucleotides that have resulted from replication errors. Researchers highlighted the importance of such repair enzymes when they found that a hereditary defect in one of them is associated with a form of colon cancer. Apparently, this defect allows cancer-causing errors to accumulate in the DNA faster than normal.

Incorrectly paired or altered nucleotides can also arise after replication. In fact, maintenance of the genetic information encoded in DNA requires frequent repair of various kinds of damage to existing DNA. DNA molecules are constantly subjected to potentially harmful chemical and physical agents, such as X-rays, as we'll discuss in Concept 17.5. In addition, DNA bases may undergo spontaneous chemical changes under normal cellular conditions. However, these changes in DNA are usually corrected before they become permanent changes—*mutations*—perpetuated through successive replications. Each cell continuously monitors and repairs its genetic material. Because repair of damaged DNA is so important to the survival of an organism, it is no surprise that many different DNA repair enzymes have evolved. Almost 100 are known in *E. coli*, and about 170 have been identified so far in humans.

Most cellular systems for repairing incorrectly paired nucleotides, whether they are due to DNA damage or to replication errors, use a mechanism that takes advantage of the base-paired structure of DNA. In many cases, a segment of the strand containing the damage is cut out (excised) by a DNA-cutting enzyme—a **nuclease**—and the resulting gap is then filled in with nucleotides, using the undamaged strand as a template. The enzymes involved in filling the gap are a DNA polymerase and DNA ligase. There are several such DNA repair systems; one is called **nucleotide excision repair**.

An example is shown in **Figure 16.20**. An important function of the DNA repair enzymes in our skin cells is to repair genetic damage caused by the UV rays of sunlight: For example, adjacent thymine bases on a DNA strand can become covalently linked into *thymine dimers*, causing the DNA to buckle and interfere with DNA replication. The importance of repairing this kind of damage is underscored by a disorder called xeroderma pigmentosum (XP), which in most cases is caused by an inherited defect in a nucleotide excision repair enzyme. Individuals with XP are hypersensitive to sunlight; mutations in their skin cells caused by ultraviolet light are left uncorrected, often resulting in skin cancer. The effects are extreme: Without sun protection, children who have XP can develop skin cancer by age 10.

Figure 16.20

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Nucleotide excision repair of DNA damage.
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Nucleotide excision repair of D N A damage.

1. Teams of enzymes detect and repair damaged D N A, such as this thymine dimer (often caused by ultraviolet radiation), which distorts the D N A molecule. Thymine dimer is two adjacent thymine bases in one D N A strand that cause a buckle and bulge in the strand.
2. A nuclease enzyme cuts the damaged D N A strand at two points, and the damaged section is removed.

3. Repair synthesis by a D N A polymerase fills in the missing nucleotides, using the undamaged strand as a template.
4. D N A ligase seals the free end of the new D N A to the old D N A, making the strand complete.

Evolutionary Significance of Altered DNA Nucleotides

EVOLUTION Faithful replication of the genome and repair of DNA damage are important for the functioning of the organism and for passing on a complete, accurate genome to the next generation. The error rate after proofreading and repair is extremely low, but rare mistakes do slip through. Once a mismatched nucleotide pair is replicated, the sequence change is permanent in the daughter molecule that has the incorrect nucleotide as well as in any subsequent copies. As we mentioned earlier, a permanent change in the DNA sequence is called a mutation.

Mutations can change the phenotype of an organism (see Concept 17.5). And if they occur in germ cells, which give rise to gametes, mutations can be passed on from generation to generation. The vast majority of such changes either have no effect or are harmful, but a very small percentage can be beneficial. In either case, mutations are the original source of the variation on which natural selection operates during evolution and are ultimately responsible for the appearance of new species. (You'll learn more about this process in Unit Four.) The balance between complete fidelity of DNA replication or repair and a low mutation rate has resulted in new proteins that contribute to different phenotypes. Ultimately, over long periods of time, this process leads to new species and thus to the rich diversity of life on Earth today.

Replicating the Ends of DNA Molecules

For linear DNA, such as the DNA of eukaryotic chromosomes, the usual replication machinery cannot complete the 5′ ends of daughter DNA strands because there is no 3′ end of a preexisting polynucleotide for DNA polymerase to add onto. This is another consequence of the enzyme's requirements. Even if an Okazaki fragment can be started with an RNA primer hydrogen-bonded to the very end of the template strand, once that primer is removed, it cannot be replaced with DNA because there is no 3′ end available for nucleotide addition **(Figure 16.21)**. As a result, repeated rounds of replication produce shorter and shorter DNA molecules with uneven ("staggered") ends.

Figure 16.21

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Shortening of the ends of linear DNA molecules.
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Diagram zooms in on lagging strand of unwound parental DNA molecule. RNA primer is removed but not replaced at the 3 prime end. Result is shorter and shorter daughter molecules.

Here we follow the left end of one DNA molecule through two rounds of replication. After the first round, the new lagging strand is shorter than its template. After a second round, both the leading and lagging strands have become shorter than the original parental DNA. Although not shown here, the other ends of these chromosomal DNA molecules (not shown) also become shorter.

Unwound parental D N A molecule with leading and lagging strands shown. Arrows zoom in on section of lagging strand at the 3 prime end of the parental template strand. The parental template runs (left to right) 3 prime to 5 prime and the lagging strand 5 prime to 3 prime. Sections of R N A primer are shown on the template between the fragments. Arrows indicate primers are removed and replaced with D N A where a 3 prime end is available. At the final 5 prime end of the lagging strand (closest to the replication fork) D N A can't be added since this is a 5 prime end and there is no 3 prime end available for D N A polymerase. Arrows indicate a second round of replication, showing a shortened new leading strand and the shortened new strand made from it. The new lagging strand is shown as original length for the (left to right) 3 prime to 5 prime strand, but the new lagging strand (5 prime to 3 prime) is shortened by the length of the previous R N A primer. Arrows indicate that following further rounds of replication, daughter molecules will become shorter and shorter.

Most prokaryotes have a circular chromosome, with no ends, so the shortening of DNA does not occur. But what protects the genes of linear eukaryotic chromosomes from being eroded away during successive rounds of DNA replication? Eukaryotic chromosomal DNA molecules have special nucleotide sequences called **telomeres** at their ends **(Figure 16.22)**. Telomeres do not contain genes; instead, the DNA typically consists of multiple repetitions of one short nucleotide sequence. In each human telomere, for example, the six-nucleotide sequence TTAGGG is repeated between 100 and 1,000 times.

Figure 16.22

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Telomeres.
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Image of mouse chromosomes in blue with telomeres visible as short bright spots on the ends. Scale: 1 micrometer.

Eukaryotes have repetitive, noncoding sequences called telomeres at the ends of their DNA. Telomeres are stained orange in these mouse chromosomes (LM).

Telomeres have two protective functions. First, specific proteins associated with telomeric DNA prevent the staggered ends of the daughter molecule from activating the cell's systems for monitoring DNA damage.

(Staggered ends of a DNA molecule, which often result from double-strand breaks, can trigger signal transduction pathways leading to cell cycle arrest or cell death.) Second, telomeric DNA acts as a kind of buffer zone that provides some protection against the organism's genes shortening, somewhat like how the plastic-wrapped ends of a shoelace slow down its unraveling. Telomeres do not prevent the erosion of genes near the ends of chromosomes; they merely postpone it.

As shown in Figure 16.21, telomeres become shorter during every round of replication. Thus, as expected, telomeric DNA tends to be shorter in dividing somatic cells of older individuals and in cultured cells that have divided many times. It has been proposed that shortening of telomeres is somehow connected to the aging process of certain tissues and even to aging of the organism as a whole.

But what about germ cells, whose genome must persist virtually unchanged from an organism to its offspring over many generations? If the chromosomes of germ cells became shorter in every cell cycle, essential genes would eventually be missing from the gametes they produce. However, this does not occur: An enzyme called *telomerase* catalyzes the lengthening of telomeres in eukaryotic germ cells, thus restoring their original length and compensating for the shortening that occurs during DNA replication. This enzyme contains its own RNA molecule that it uses as a template to artificially "extend" the leading strand, allowing the lagging strand to maintain a given length. Telomerase is not active in most human somatic cells, but its activity varies from tissue to tissue. The activity of telomerase in germ cells results in telomeres of maximum length in the zygote.

BBC Video

Can Humans Live Forever?

Normal shortening of telomeres may protect organisms from cancer by limiting the number of divisions that somatic cells can undergo. Cells from large tumors often have unusually short telomeres, as we would expect for cells that have undergone many cell divisions. Further shortening would presumably lead to self-destruction of the tumor cells. Telomerase activity is abnormally high in cancerous somatic cells, suggesting that its ability to stabilize telomere length may allow these cancer cells to persist. Many cancer cells do seem capable of unlimited cell division, as do immortal strains of cultured cells (see Concept 12.3). For several years, researchers have studied inhibition of telomerase as a possible cancer therapy. While studies that inhibited telomerase in mice with tumors have led to the death of cancer cells, eventually the cells have restored the length of their telomeres by an alternative pathway. This is an area of ongoing research that may eventually yield useful cancer treatments.

J. D. Watson and F. H. C. Crick, Molecular structure of nucleic acids: a structure for deoxyribose nucleic acids, *Nature* 171:737–738 (1953).

J. D. Watson and F. H. C. Crick, Genetical implications of the structure of deoxyribonucleic acid, *Nature* 171:964–967 (1953).

Concept 16.3: A chromosome consists of a DNA molecule packed together with proteins

Now let's examine how DNA is packaged into chromosomes, the structures that carry genetic information. The main component of the genome in most bacteria is a double-stranded, circular DNA molecule that is associated with specific proteins. A bacterial chromosome differs from a eukaryotic chromosome in that a eukaryotic chromosome consists of a single linear DNA molecule associated with a large number of proteins. In *E. coli*, the chromosomal DNA consists of about 4.6 million nucleotide pairs, representing about 4,400 genes. This is 100 times more DNA than is found in a typical virus, but only about one-thousandth as much DNA as in a human somatic cell. Even so, that is a tremendous amount of DNA to be packaged in such a small container.

Stretched out, the DNA of an *E. coli* cell would measure about a millimeter in length, 1,000 times longer than the region it occupies in the cell. Within a bacterium, however, certain proteins cause the chromosome to coil and "supercoil," densely packing it so that it fills only part of the cell. Unlike the nucleus of a eukaryotic cell, this dense region of DNA in a bacterium, called the nucleoid, is not bounded by membrane (see Figure 6.5).

Each eukaryotic chromosome contains a single linear DNA double helix that, in humans, averages about 1.5×108 nucleotide pairs. This is an enormous amount of DNA relative to a chromosome's condensed length. If completely stretched out, such a DNA molecule would be about 4 cm long, thousands of times the diameter of a cell nucleus—and that's not even considering the DNA of the other 45 human chromosomes!

In the cell, eukaryotic DNA is precisely combined with a large amount of protein. The exact way in which this complex of DNA and protein, called ****__chromatin__, fits into the nucleus has long been debated. **Figure 16.23** outlines the current view of the organization of chromatin during interphase and how it is condensed during mitosis into the metaphase chromosome. Study this figure carefully before reading further.

Figure 16.23

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Exploring Chromatin Packing in a Eukaryotic Chromosome
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Exploring chromatin packing in a eukaryotic chromosome.

D N A, the double helix.

T E M shows D N A appearing as thin line of segments winding through a dotted area. Shown is a ribbon model of D N A, with each ribbon representing one of the polynucleotide strands. Each phosphate group along the backbone of the strand contributes a negative charge along the outside of each strand. The T E M shows a molecule of naked (protein-free) D N A; the double helix alone is 2 nanometers across and 2 nanometers in diameter.

Histones (drawn beside the ribbon model as spheres with tails). Proteins called histones are responsible for the first level of D N A packing in chromatin. More than a fifth of a histone's amino acids are positively charged (lysine or arginine) and therefore bind tightly to the negatively charged D N A (caused by the negatively charged phosphate groups in the backbone of the D N A strand). Four types of histones are most common in chromatin. The histones are very similar among eukaryotes; probably reflecting their important role in chromatin structure.

Nucleosomes in a 10-nanometer fiber (beads on a string (10-nanometer fiber). In the diagram, nucleosomes are depicted as beads on a string of D N A (T E M). The D N A from the ribbon model winds up and around 8 histone proteins, with spheres inside the beads and tails sticking out. In electron micrographs, unfolded chromatin is 10 nanometers in diameter (euchromatin loosely arranged 10-nanometer fiber and 10 nanometers in diameter). Such chromatin resembles beads on a string (see the T E M). Each "bead" is a nucleosome, the basic unit of D N A packing; the "string" between beads is called linker D N A. A nucleosome consists of D N A wound twice around a protein core of eight histones, two each of the four main histone types. The amino end (N-terminus) of each histone (the histone tail) extends outward from the nucleosome and is involved in regulation of gene expression. In the cell cycle, the histones leave the D N A only briefly during D N A replication. Generally, they do the same during the process of transcription, which requires access to the D N A by transcription process. Nucleosomes, and in particular their histone tails, are involved in the regulation of gene expression.

Euchromatin/Heterochromatin. Many beads (nucleosomes) arranged on a network of strings of D N A is illustrated in a diagram labeled heterochromatin (densely arranged 10-nanometer fiber). A recent technique called Chrom E M T allows scientists to visualize chromatin in intact cells. Chrom E M T and other new techniques have shown that the 10-nanometer fiber is the basic constituent of interphase chromatin. During interphase, different regions of a chromosome may exist as euchromatin (below left) or heterochromatin (below). In euchromatin, the 10-nanometer fiber is loosely arranged in a more open configuration than heterochromatin; higher degrees of organization, including the 30-nanometer fiber in one older model, may exist in specific cells or at certain times. (This is an area of very active research.) The D N A in euchromatin is accessible to the proteins that carry out transcription, and its genes can be expressed. In heterochromatin,

the 10-nanometer fiber is more densely arranged and less accessible to these proteins; genes in heterochromatin are generally not expressed. Euchromatin and heterochromatin are organized into regions by other proteins not shown here; this organization is dynamic but disappears once mitosis begins.

Figure 16.23 continues with prophase, prometaphase, and metaphase.

Prophase. When mitosis begins, D N A replication has already occurred, so each chromosome consists of two sister chromatids. During prophase, the chromatin of each sister chromatid begins to condense. Two related proteins called condensin Roman numeral 2 and condensin Roman numeral 1 play important roles. First, condensin Roman numeral 2 proteins (red in the diagram of a string of red beads with loops connecting them) bind to the 10-n m fiber of D N A and form D N A loops that get larger and larger. The condensin Roman numeral 2 proteins form a central scaffold from which the loops extend. As the loops grow, the chromosome gets wider and shorter. By the end of prophase, the chromosome is half as long.

Prometaphase. When prometaphase begins, condensin Roman numeral 1 proteins (green in the diagram of the larger loops) bind to D N A outside the central scaffold, making smaller loops (not shown) out of the larger loops generated by condensin Roman numeral 2. The process continues, with more and more loops extending outward, and the chromosome getting denser, shorter, and wider. The scaffold itself also begins to twist into a helix (suggested by the curved gray arrows around the central scaffold with loops), allowing even more loops per given length of chromosome.

Metaphase. At metaphase, the chromosome is at its most dense, with the most loops per turn, and therefore is at its shortest length. The two sister chromatids are fully condensed. An enlarged portion of a chromosome is a diagram tat shows the sister chromatid having a diameter of 700 nanometers.

Animation

DNA Packing

image

Play

image

Play

Chromatin undergoes striking changes in how densely it is packed during the course of the cell cycle (see Figure 12.7). In interphase cells stained for light microscopy, the chromatin usually appears as a diffuse mass within the nucleus, with some denser clumps, including in regions of centromeres and telomeres. The less compacted, more dispersed interphase chromatin is called **euchromatin** (“true chromatin”) to distinguish it from the more compacted, denser-appearing **heterochromatin** (see Figure 16.23). Recent research has

clarified our understanding of chromatin structure: For both types of chromatin, the basic organizing unit is the 10-nm fiber—nucleosomes joined by linker DNA. In heterochromatin, this 10-nm fiber is folded and bent back on itself to a much greater degree than in euchromatin, accounting for its denser appearance. Because heterochromatin is so compacted, it is largely inaccessible to the proteins responsible for transcribing the genetic information, a crucial early step in gene expression. In contrast, the looser packing of euchromatin makes its DNA accessible to those proteins, and the genes present in euchromatin are available for transcription.

Early on, biologists assumed that interphase chromatin was a tangled mass in the nucleus, like a bowl of spaghetti, but this is far from the case. Although an interphase chromosome lacks an obvious scaffold, there are proteins that further organize the 10-nm fiber into larger compartments and smaller looped domains. Some of the looped domains appear to be attached to the nuclear lamina, on the inside of the nuclear envelope, and perhaps also to fibers of the nuclear matrix. These attachments may help organize regions of chromatin where genes are active. The chromatin of each chromosome occupies a specific restricted area within the interphase nucleus, and the chromatin fibers of different chromosomes do not appear to be entangled **(Figure 16.24a)**.

Figure 16.24

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“Painting” chromosomes.
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Painting chromosomes.

Researchers can treat (“paint”) human chromosomes with molecular tags that cause each chromosome pair to appear a different color.

Part a. Image shows diffuse splotches of color in the interphase nucleus. Similar colors are not adjacent. Scale: 5 micrometers. The ability to visually distinguish among chromosomes makes it possible to see how the chromosomes are arranged in the interphase nucleus. Each chromosome appears to occupy a specific territory during interphase. In general, the two homologs of a pair are not located together.

Part b. Scatter of metaphase chromosomes appear as thick lines; homologous pairs are the same color and are organized into a karyotype by pair, showing 22 numbered pairs and two X chromosomes. These metaphase chromosomes have been painted so that the two homologs of a pair are the same color. Top image shows a spread of treated chromosomes. Image on the right shows them organized into a karyotype.

MAKE CONNECTIONS*If you arrested a human cell in metaphase I of meiosis and applied this technique, what would you observe? How would this differ from what you would see in metaphase of mitosis? Review***Figure 13.8***and*Figure 12.7.**

*For suggested answer, see*Appendix A.**

As a cell prepares for mitosis, its chromatin becomes organized into loops and coils, eventually condensing into a characteristic number of short, thick metaphase chromosomes that are distinguishable from each other with the light microscope **(Figure 16.24b)**.

The chromosome is a dynamic structure that is condensed, loosened, modified, and remodeled as necessary for various cell processes, including DNA replication, mitosis, meiosis, and gene expression. Certain chemical modifications of histones affect the state of chromatin condensation and also have multiple effects on gene expression, as you'll see in Concept 18.2.

In this chapter, you have learned how DNA molecules are arranged in chromosomes and how DNA replication provides the copies of genes that parents pass to offspring. However, it is not enough that genes be copied and transmitted; the information they carry must be used by the cell. In other words, genes must also be expressed. In the next chapter, we will examine how the cell expresses the genetic information encoded in DNA.